SR7000
TS2000
TS2100
Robot Controller

I/O cables MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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Preface

This manual describes the I/O cables used for industrial robot controller SR7000/TS2000/TS2100. These I/O cables are optional. For use of the cables, see the Interface Manual or I/O common Manual (type p).



CAUTION

This manual does not include detailed descriptions on connecting the robot, which are contained in the Installation & Transport Manual and Interface Manual or I/O common Manual (type p).

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2).

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.



CAUTION

To assure the safety work ranging from the robot installation to operation, read through the Safety Manual before starting the work.

[Maintenance and inspection]

To use the robot safely, strictly observe the following matters.

 **DANGER**	
Mandatory	• Before starting the maintenance and inspection, be sure to turn off the main power switch of the controller.

 **CAUTION**	
Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the robot performance will deteriorate or faults will be caused.
Mandatory	• Always use the spare parts designated by Toshiba Machine when replacing the parts. • Service and inspect the robot on a regular basis. Otherwise, the robot may malfunction or cause an accident as a result.

 **CAUTION**

To assure the safety work ranging from the robot installation to operation, read through the Safety Manual before starting the work.

1. CN5 Cable for SR7000/TS2000/TS2100 Controller

1.1 Specifications

Type : 368M04□□
Cable length : L=__m

＊ For details and connection of the input signals, see the Interface Manual (SR7000:STE58763 / TS2000,TS2100:STE71367) or I/O common Manual (type p) (STE71380).

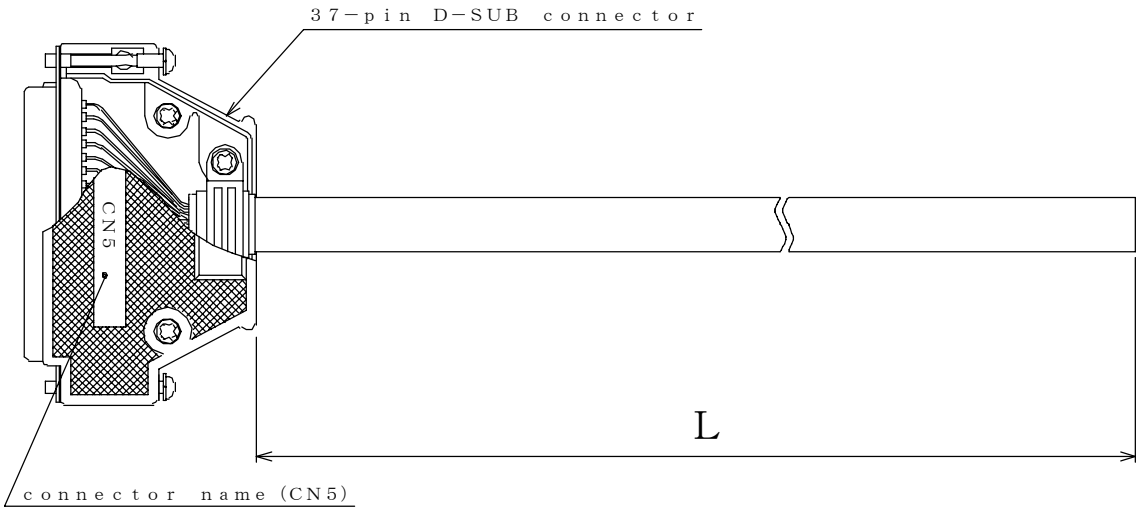
Cable specifications

Items	Specifications
Nominal sectional area of conductor	0.2 mm^2
Structure of conductor	7/0.18
No. of cable cores	40C
Insulator material	Heat-proof vinyl chloride
Shield material	Tin-plated annealed copper wire
Sheath material	Heat-proof vinyl chloride
Conductor resistance (20°C)	119 Ω/km or less
Operating temperature	–20°C ~ 75°C (See Note 1.)
Operating voltage	AC 50 V, DC 60 V

Note 1: Heat-proof grade: 90°C

Note 2: This cable cannot be used for the sliding part or moving part which will bend repeatedly.

1.2 External View



1.3 Cable Connection Table

Signal name		Connector pin	Cable color
SR7000	TS2000/TS2100		
DI1	DIN1	1	Black
DI2	DIN2	20	White
DI3	DIN3	2	Red
DI4	DIN4	21	Green
DI5	DIN5	3	Yellow
DI6	DIN6	22	Brown
DI7	DIN7	4	Blue
DI8	DIN8	23	Gray
DI9	DIN9	5	Orange
DI10	DIN10	24	Purple
DI11	DIN11	6	Peach
DI12	DIN12	25	Light green
DI13	DIN13	7	Light blue
DI14	DIN14	26	Black／White
DI15	DIN15	8	Red／White
DI16	DIN16	27	Green／White
DI17	DIN17	9	Yellow／White
DI18	DIN18	28	Brown／White
DI19	DIN19	10	Blue／White
DI20	DIN20	29	Gray／White
DI21	DIN21	11	Orange／White
DI22	DIN22	30	Purple／White
DI23	DIN23	12	Peach／White
DI24	DIN24／ALM_RST	31	Light green／White
Strobe	STROBE／DIN33	13	Light blue／White
Program reset	PRG_RST／DIN34	32	White／Black
Step reset	STEP_RST／DIN35	14	Red／Black
Cycle reset	CYC_RST／DIN36	33	Green／Black
Output signal reset	DO_RST／DIN37	15	Yellow／Black
Start	RUN／DIN38	34	Brown／Black
Stop	STOP	16	Blue／Black
Cycle operation mode	CYCLE	35	Gray／Black
	BREAK	17	Orange／Black
Low speed command	LOW_SPD	36	Purple／Black
PG	P24G (P24V)	18	Peach／Black
PG	P24G (P24V)	37	Light green／Black
PG	P24G (P24V)	19	Light blue／Black
FG	FG	Case	Shield

2. CN6 Cable for the SR7000/TS2000/TS2100 Controller

2.1 Specifications

Type : 368M05□□
Cable length : L=__m

＊ For details and connection of the output signals, see the Interface Manual (SR7000:STE58763 / TS2000,TS2100:STE71367) or I/O common Manual (type p) (STE71380).

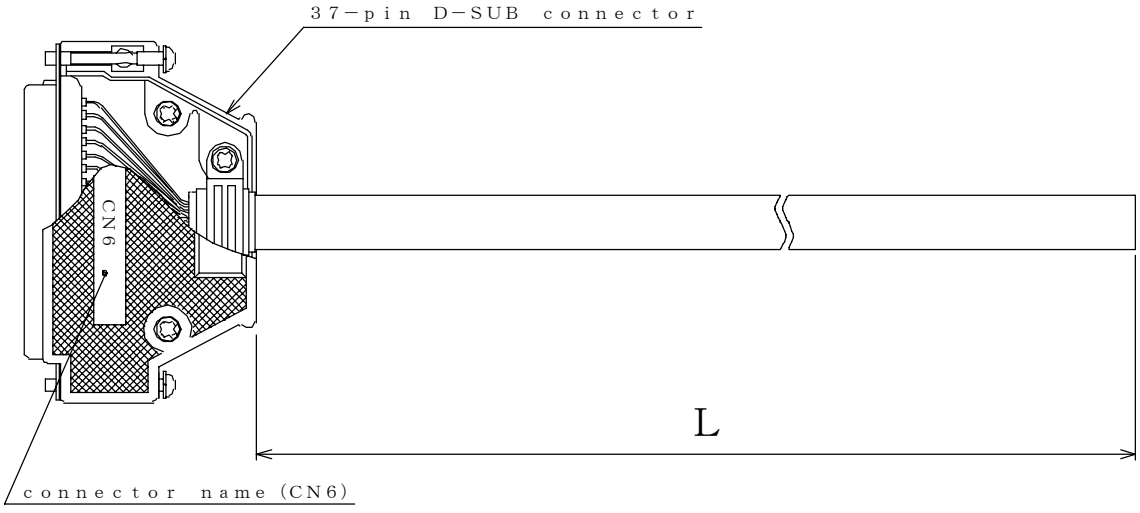
Cable specifications

Items	Specifications
Nominal sectional area of conductor	0.2 mm^2
Structure of conductor	7/0.18
No. of cable cores	40C
Insulator material	Heat-proof vinyl chloride
Shield material	Tin-plated annealed copper wire
Sheath material	Heat-proof vinyl chloride
Conductor resistance (20°C)	119 Ω/km or less
Operating temperature	–20°C ~ 75°C (See Note 1.)
Operating voltage	AC 50 V, DC 60 V

Note 1: Heat-proof grade: 90°C

Note 2: This cable cannot be used for the sliding part or moving part which will bend repeatedly.

2.2 External View



2.3 Cable Connection Table

Signal name		Connector pin	Cable color
SR7000	TS2000/TS2100		
DO1	DOUT1	1	Black
DO2	DOUT2	20	White
DO3	DOUT3	2	Red
DO4	DOUT4	21	Green
DO5	DOUT5	3	Yellow
DO6	DOUT6	22	Brown
DO7	DOUT7	4	Blue
DO8	DOUT8	23	Gray
DO9	DOUT9	5	Orange
DO10	DOUT10	24	Purple
DO11	DOUT11	6	Peach
DO12	DOUT12	25	Light green
DO13	DOUT13	7	Light blue
DO14	DOUT14	26	Black／White
DO15	DOUT15／SV_RDY	8	Red／White
DO16	DOUT16／BT_ALM	27	Green／White
Acknowledge	ACK／DOUT25	9	Yellow／White
In manual mode	TEACH／DOUT26	28	Brown／White
In external automatic mode	EXTSIG／DOUT27	10	Blue／White
Ready for operation	SYS_RDY／DOUT28	29	Gray／White
In automatic operation	AUTORUN／DOUT29	11	Orange／White
Fault	ALARM／DOUT30	30	Purple／White
Cycle end	CYC_END／DOUT31	12	Peach／White
In low speed mode	LOW_ST／DOUT32	31	Light green／White
In servo on	SVST_A	13	Light blue／White
	SVST_B	32	White／Black
In emergency stop	EMSST_A	14	Red／Black
	EMSST_B	33	Green／Black
Servo on	EX_SVON	15	Yellow／Black
	SVOFF	34	Brown／Black
Servo off	EMS2B	16	Blue／Black
	EMS2C	35	Gray／Black
Emergency stop	EMS1B	17	Orange／Black
	EMS1C	36	Purple／Black
P24V	P24V (P24G)	18	Peach／Black
P24V	P24V (P24G)	37	Light green／Black
P24V	P24V (P24G)	19	Light blue／Black
FG	FG	Case	Shield

3. CN12 Cable for the SR7000/TS2000/TS2100 Controller

3.1 Specifications

Type : 368M06□□
Cable length : L=__m

∗ For details and connection of the input and output signals, see the Interface Manual (SR7000:STE58763 / TS2000,TS2100:STE71367) or I/O common Manual (type p) (STE71380).

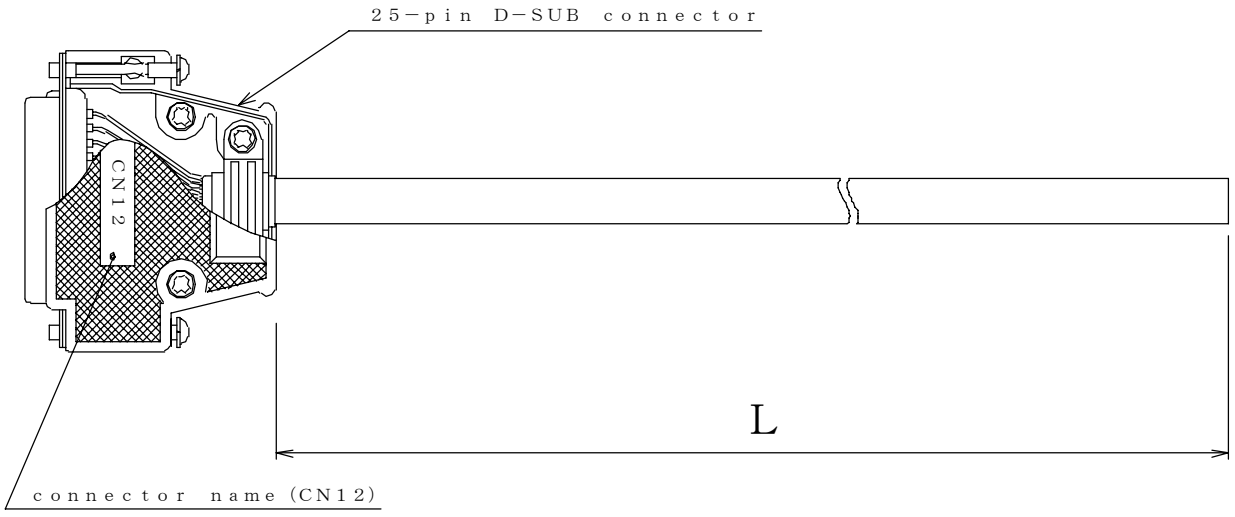
Cable specifications

Items	Specifications
Nominal sectional area of conductor	0.2 mm^2
Structure of conductor	7/0.18
No. of cable cores	20C
Insulator material	Heat-proof vinyl chloride
Shield material	Tin-plated annealed copper wire
Sheath material	Heat-proof vinyl chloride
Conductor resistance (20°C)	119 Ω/km or less
Operating temperature	–20°C ~ 75°C (See Note 1.)
Operating voltage	AC 50 V, DC 60 V

Note 1: Heat-proof grade: 90°C

Note 2: This cable cannot be used for the sliding part or moving part which will bend repeatedly.

3.2 External View



3.3 Cable Connection Table

Signal name		Connector pin	Cable color
SR7000	TS2000/TS2100		
DI25	DIN25	1	Black
DI26	DIN26	14	White
DI27	DIN27	2	Red
DI28	DIN28	15	Green
DI29	DIN29	3	Yellow
DI30	DIN30	16	Brown
DI31	DIN31	4	Blue
DI32	DIN32	17	Gray
P24V	P24V (P24G)	5	Orange
P24V	P24V (P24G)	18	Purple
DO17	DOUT17	6	Peach
DO18	DOUT18	19	Light green
DO19	DOUT19	7	Light blue
DO20	DOUT20	20	Black／White
DO21	DOUT21	8	Red／White
DO22	DOUT22	21	Green／White
DO23	DOUT23	9	Yellow／White
DO24	DOUT24	22	Brown／White
PG	P24G (P24V)	10	Blue／White
PG	P24G (P24V)	23	Gray／White
		11	
		24	
		12	
		25	
		13	
FG	FG	Case	Shield

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